

Basic Information

Experiment No.: 07

Date: 24-09-2026

Title: Java LinkedHashSet and Its Methods

Objective / Aim

To implement "LinkedHashSet" in Java and demonstrate insertion, duplicate handling, removal, searching, size checking, and clearing elements.

System Requirements

Hardware

- Computer/Laptop
- Minimum 4 GB RAM

Software

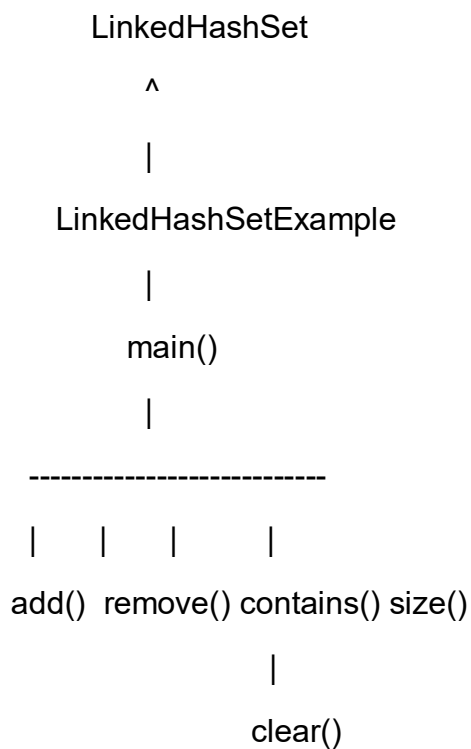
- Java JDK
 - Any Java IDE or Command Prompt
 - Operating System: Windows/Linux/macOS
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Algorithm / Logic Description

1. Create a "LinkedHashSet" of strings.
2. Add elements using "add()".
3. Add a duplicate element and observe that it is not stored.
4. Display the elements in insertion order.
5. Remove an element using "remove()".

6. Check an element using "contains()".
 7. Find the number of elements using "size()".
 8. Remove all elements using "clear()".
 9. End the program.
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UML Class Diagram



Source Code

```
package collectionframework;

import java.util.LinkedHashSet;

public class LinkedHashSetExample {

    public static void main(String[] args) {

        LinkedHashSet<String> set = new LinkedHashSet<>();
```

```
// add()
set.add("Apple");
set.add("Banana");
set.add("Mango");
set.add("Apple"); // Duplicate

System.out.println("LinkedHashSet: " + set);

// remove()
set.remove("Banana");
System.out.println("After remove: " + set);

// contains()
System.out.println("Contains Apple: " +
    set.contains("Apple"));

// size()
System.out.println("Size: " + set.size());

// clear()
set.clear();

System.out.println("After clear: " + set);
}
}
```

Compilation & Execution Commands

Save the file as:

LinkedHashSetExample.java

Since the program contains:

package collectionframework;

the file should be inside the "collectionframework" folder.

Compile:

javac LinkedHashSetExample.java

Run:

java collectionframework.LinkedHashSetExample

Output Screen

LinkedHashSet: [Apple, Banana, Mango]

After remove: [Apple, Mango]

Contains Apple: true

Size: 2

After clear: []

Result

The "LinkedHashSet" class was successfully implemented, and operations such as "add()", "remove()", "contains()", "size()", and "clear()" were successfully demonstrated. Duplicate elements were not stored, and insertion order was maintained.